SEMiX453GB12E4s



SEMiX® 3s

Trench IGBT Modules

SEMiX453GB12E4s

Features

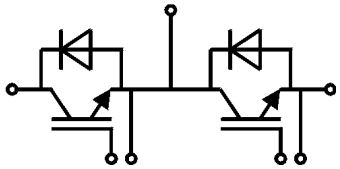
- Homogeneous Si
- Trench = Trenchgate technology
- $V_{CE(sat)}$ with positive temperature coefficient
- High short circuit capability
- UL recognized, file no. E63532

Typical Applications*

- AC inverter drives
- UPS
- Electronic Welding

Remarks

- Case temperature limited to T_C=125°C max.
- Product reliability results are valid for T_j=150°C
- Dynamic values apply to the following combination of resistors:
  $R_{Gon,main}$ = 1,0 Ω
  $R_{Goff,main}$ = 1,0 Ω
  $R_{G,X}$ = 2,2 Ω
  $R_{E,X}$ = 0,5 Ω
- For storage and case temperature with TIM see document "TP(*) SEMiX 3s"



GB

Absolute Maximum Ratings				
Symbol	**Conditions**		**Values**	**Unit**
IGBT				
V_{CES}	T_j = 25 °C		1200	V
I_C	T_j = 175 °C	T_c = 25 °C	683	A
		T_c = 80 °C	526	A
I_{Cnom}			450	A
I_{CRM}	I_{CRM} = 3xI_{Cnom}		1350	A
V_{GES}			-20 ... 20	V
t_{psc}	V_{CC} = 800 V V_{GE} ≤ 15 V V_{CES} ≤ 1200 V	T_j = 150 °C	10	µs
T_j			-40 ... 175	°C
Inverse diode				
V_{RRM}	T_j = 25 °C		1200	V
I_F	T_j = 175 °C	T_c = 25 °C	544	A
		T_c = 80 °C	407	A
I_{Fnom}			450	A
I_{FRM}	I_{FRM} = 3xI_{Fnom}		1350	A
I_{FSM}	t_p = 10 ms, sin 180°, T_j = 25 °C		2430	A
T_j			-40 ... 175	°C
Module				
$I_{t(RMS)}$			600	A
T_{stg}	module without TIM		-40 ... 125	°C
V_{isol}	AC sinus 50Hz, t = 1 min		4000	V

Characteristics						
Symbol	**Conditions**		**min.**	**typ.**	**max.**	**Unit**
IGBT						
$V_{CE(sat)}$	I_C = 450 A V_{GE} = 15 V chiplevel	T_j = 25 °C		1.80	2.05	V
		T_j = 150 °C		2.19	2.40	V
V_{CE0}	chiplevel	T_j = 25 °C		0.8	0.9	V
		T_j = 150 °C		0.7	0.8	V
r_{CE}	V_{GE} = 15 V chiplevel	T_j = 25 °C		2.2	2.6	mΩ
		T_j = 150 °C		3.3	3.6	mΩ
$V_{GE(th)}$	V_{GE}=V_{CE}, I_C = 18 mA		5	5.8	6.5	V
I_{CES}	V_{GE} = 0 V, V_{CE} = 1200 V, T_j = 25 °C				5	mA
C_{ies}	V_{CE} = 25 V V_{GE} = 0 V	f = 1 MHz		27.9		nF
C_{oes}		f = 1 MHz		1.74		nF
C_{res}		f = 1 MHz		1.53		nF
Q_G	V_{GE} = - 8 V...+ 15 V			2550		nC
R_{Gint}	T_j = 25 °C			1.7		Ω
$t_{d(on)}$	V_{CC} = 600 V I_C = 450 A V_{GE} = +15/-15 V $R_{G\ on}$ = 1.9 Ω $R_{G\ off}$ = 1.9 Ω di/dt$_{on}$ = 4000 A/µs di/dt$_{off}$ = 5000 A/µs	T_j = 150 °C		336		ns
t_r		T_j = 150 °C		80		ns
E_{on}		T_j = 150 °C		45		mJ
$t_{d(off)}$		T_j = 150 °C		615		ns
t_f		T_j = 150 °C		130		ns
E_{off}		T_j = 150 °C		66.5		mJ
$R_{th(j-c)}$	per IGBT				0.065	K/W
$R_{th(c-s)}$	per IGBT (λ_{grease}=0.81 W/(m*K))			0.03		K/W
$R_{th(c-s)}$	per IGBT, pre-applied phase change material			0.021		K/W



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Typical Applications*

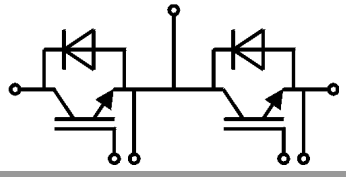
- AC inverter drives
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Remarks

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Characteristics

Symbol	Conditions		min.	typ.	max.	Unit
Inverse diode						
$V_F = V_{EC}$	I_F = 450 A V_{GE} = 0 V chiplevel	T_j = 25 °C		2.14	2.46	V
		T_j = 150 °C		2.07	2.38	V
V_{F0}	chiplevel	T_j = 25 °C		1.30	1.50	V
		T_j = 150 °C		0.90	1.10	V
r_F	chiplevel	T_j = 25 °C		1.87	2.1	mΩ
		T_j = 150 °C		2.6	2.8	mΩ
I_{RRM}	I_F = 450 A di/dt_{off} = 5000 A/µs V_{GE} = -15 V V_{CC} = 600 V	T_j = 150 °C		350		A
Q_{rr}		T_j = 150 °C		70		µC
E_{rr}		T_j = 150 °C		28		mJ
$R_{th(j-c)}$	per diode				0.11	K/W
$R_{th(c-s)}$	per diode (λ_{grease}=0.81 W/(m*K))			0.045		K/W
$R_{th(c-s)}$	per diode, pre-applied phase change material			0.036		K/W
Module						
L_{CE}				20		nH
$R_{CC'+EE'}$	measured per switch	T_C = 25 °C		0.7		mΩ
		T_C = 125 °C		1		mΩ
$Rth_{(c-s)1}$	calculated without thermal coupling			0.009		K/W
$Rth_{(c-s)2}$	including thermal coupling, Ts underneath module (λ_{grease}=0.81 W/(m*K))			0.013		K/W
$Rth_{(c-s)2}$	including thermal coupling, Ts underneath module, pre-applied phase change material			0.01		K/W
M_s	to heat sink (M5)		3		5	Nm
M_t		to terminals (M6)	2.5		5	Nm
						Nm
w					300	g
Temperature Sensor						
R_{100}	T_c=100°C (R_{25}=5 kΩ)			493 ± 5%		Ω
$B_{100/125}$	$R_{(T)}=R_{100}\exp[B_{100/125}(1/T-1/T_{100})]$; T[K];			3550 ±2%		K



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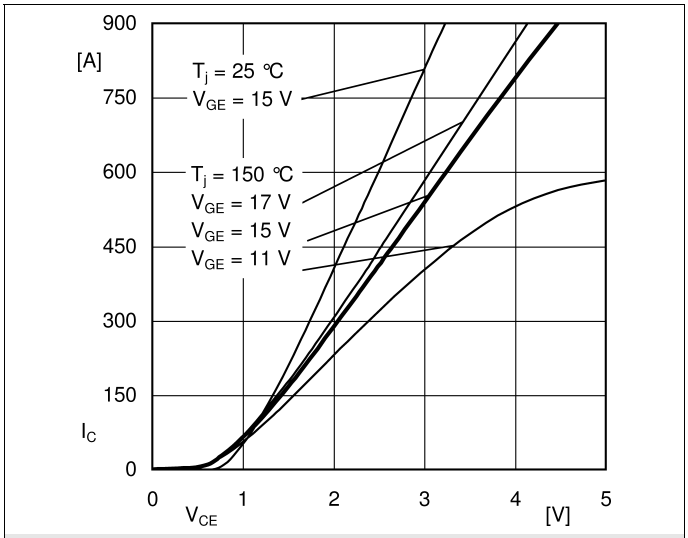


Fig. 1: Typ. output characteristic, inclusive $R_{CC'+EE'}$

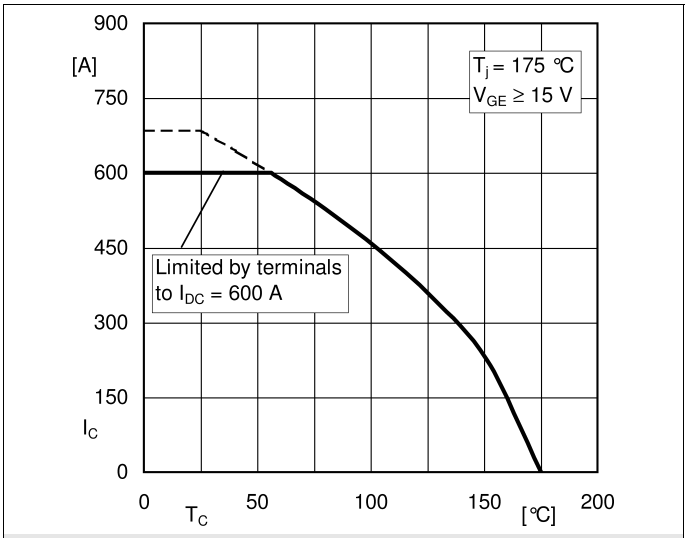


Fig. 2: Rated current vs. temperature $I_C = f(T_C)$

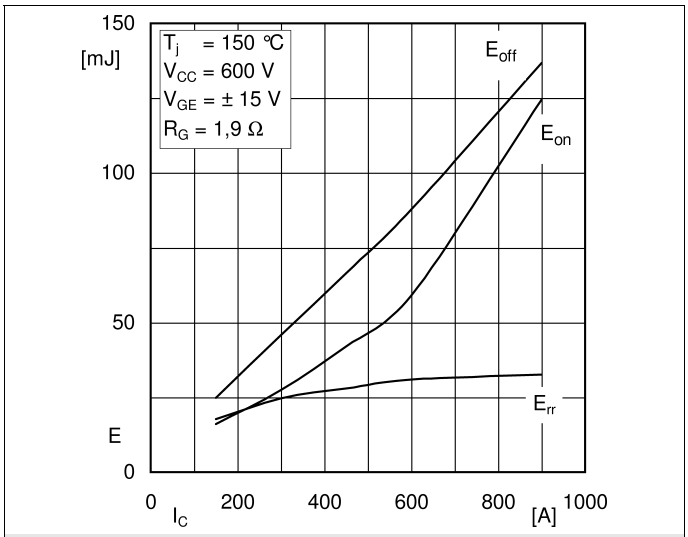


Fig. 3: Typ. turn-on /-off energy = f (I_C)

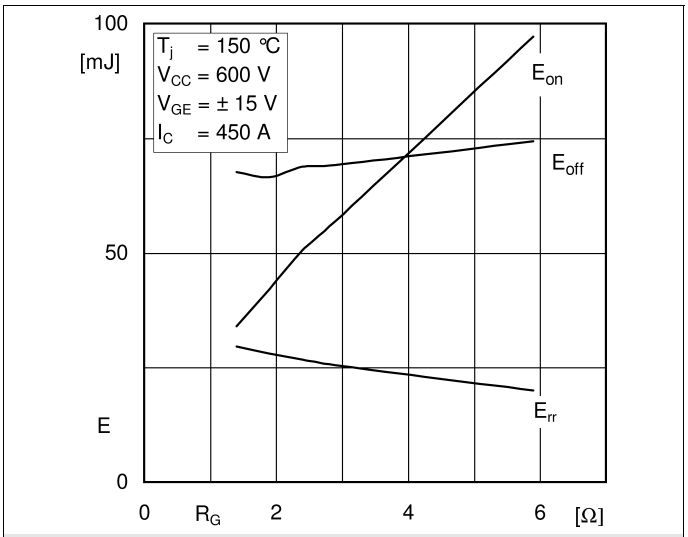


Fig. 4: Typ. turn-on /-off energy = f (R_G)

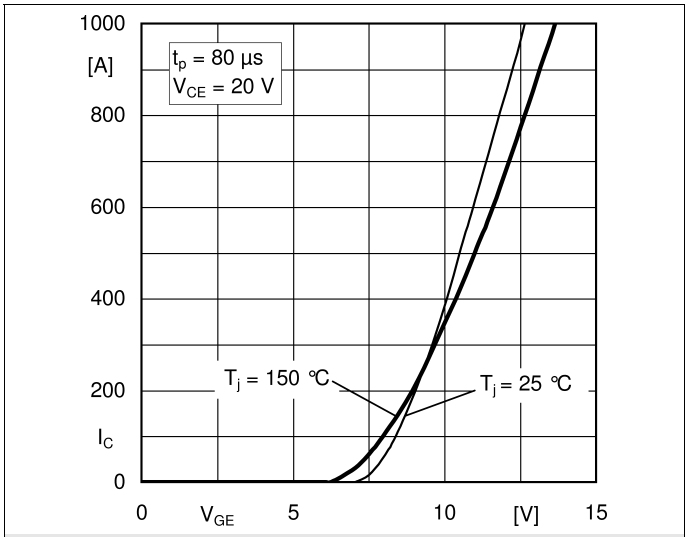


Fig. 5: Typ. transfer characteristic

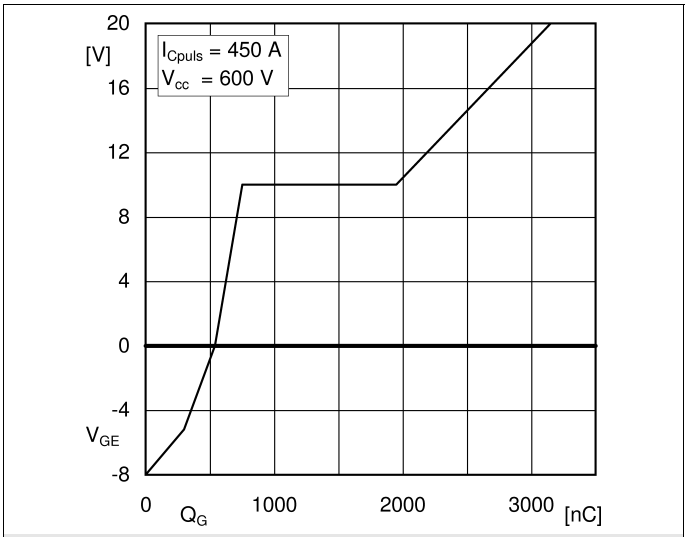


Fig. 6: Typ. gate charge characteristic

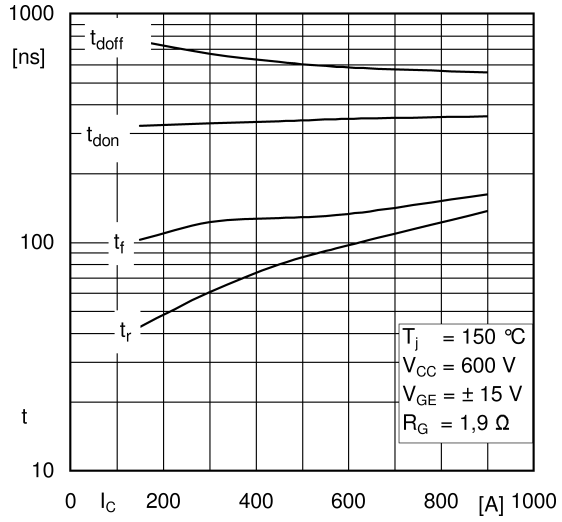


Fig. 7: Typ. switching times vs. I_C

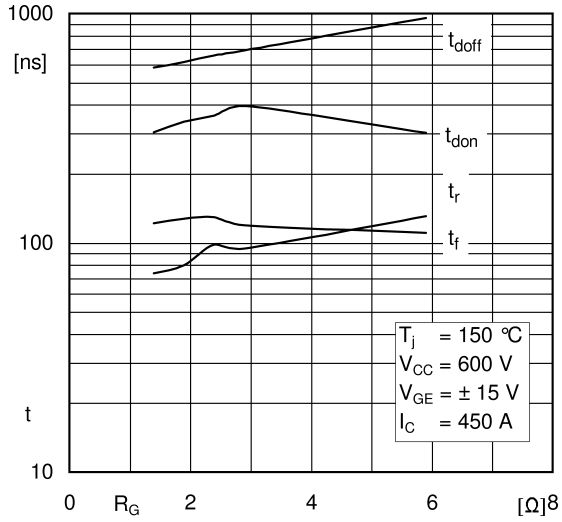


Fig. 8: Typ. switching times vs. gate resistor R_G

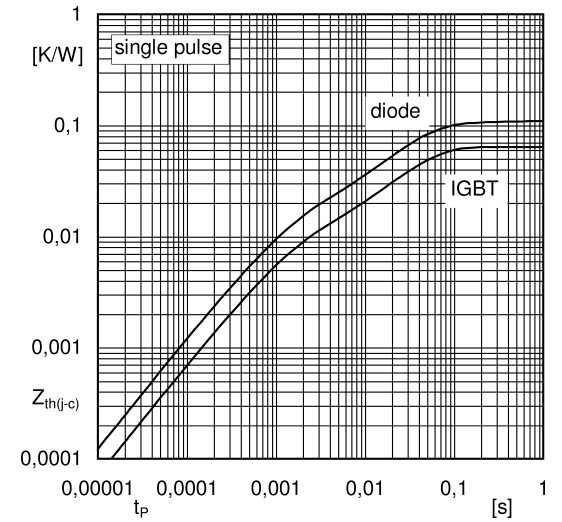


Fig. 9: Transient thermal impedance

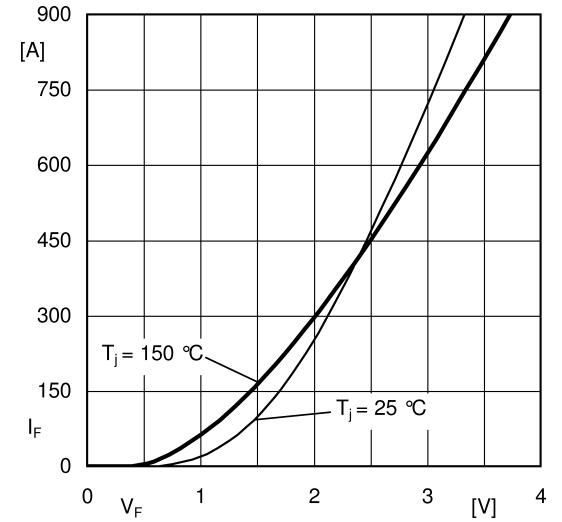


Fig. 10: Typ. CAL diode forward charact., incl. $R_{CC'+EE'}$

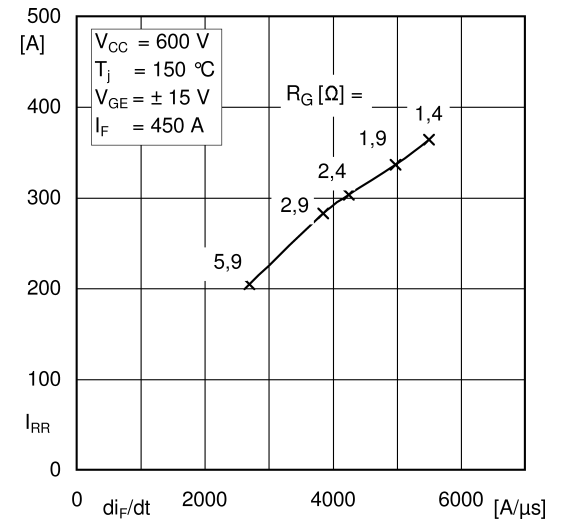


Fig. 11: Typ. CAL diode peak reverse recovery current

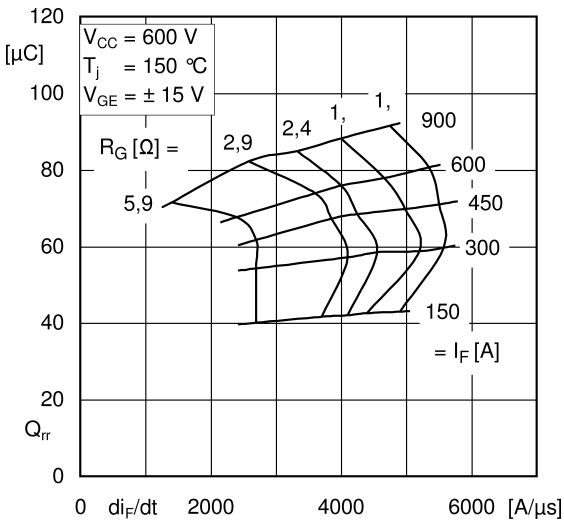
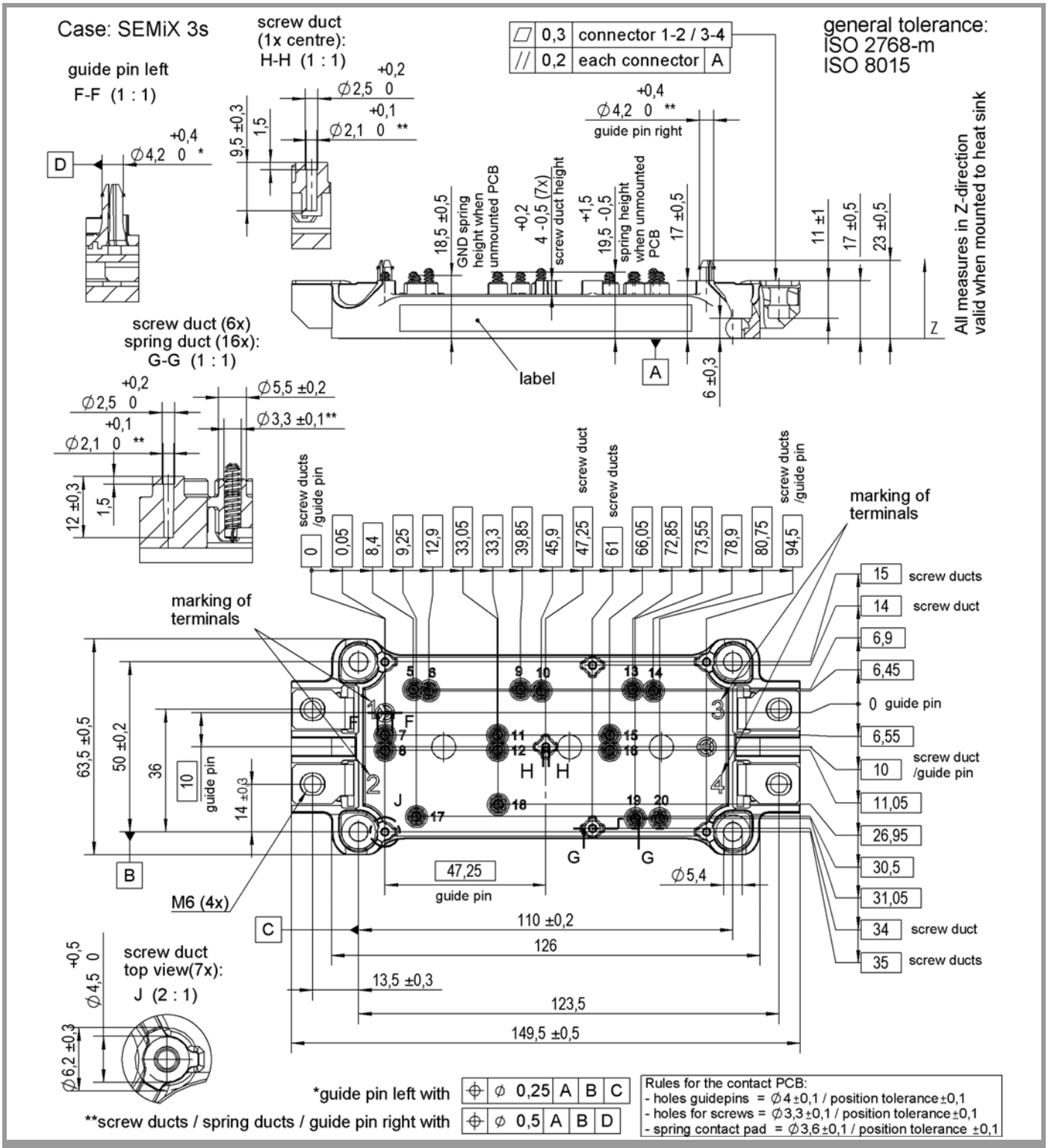
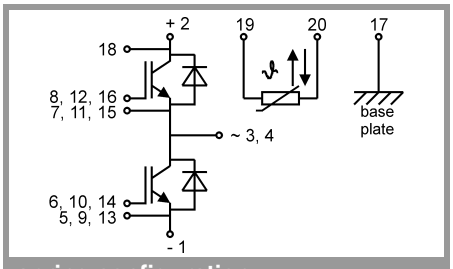


Fig. 12: Typ. CAL diode recovery charge

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spring configuration

This is an electrostatic discharge sensitive device (ESDS), international standard IEC 60747-1, chapter IX.

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